Ref. Ares(2020)1656010 - 19/03/2020

ESSnet Big Data II

Grant Agreement Number: 847375-2018-NL-BIGDATA

https://webgate.ec.europa.eu/fpfis/mwikis/essnetbigdata
https://ec.europa.eu/eurostat/cros/content/essnetbigdata_en

Workpackage WPB

Implementation – Online Job Vacancies

Methodological framework for processing online job adverts data for Official Statistics V.1.

Prepared by:
Galya Stateva, Francis Saucy, Annalisa Lucarelli, Dan Wu, Jacek Maślankowski, Claire Dumesnil de Maricourt, Črt Grahonja, Tomaž Špeh

Date of present version: 18 March 2020

Workpackage Leader:
Tomaž Špeh (SURS, SI)
e-mail address: tomaz.speh@gov.si
mobile phone: +38651672116

Contents

Purpose and scope 5
Methodological considerations around OJAs 7
Labour market main concepts 9
Job adverts main concepts used 10
Online job vacancies: seen as proxies to statistical indicators 12
Short description of the JV survey process 13
Peculiarities and weaknesses of Job Vacancy statistics 14
OJA web scraping in the context of the big data life cycle 15
Possible statistical indicators 16
Complementation of JVS 19
Choice of websites: enterprise websites versus job portals 19
Some quality aspects of OJA data 21
Coverage and selectivity 21
Measurement errors 23
Comparability over time 23
Linkability 23
Quality of individual matching in the Swiss example 24
Identifying enterprises in business registers 25
Pseudo-stocks 26
Processing errors 26
Fictional OJAs 27
ESS OJA reference production architecture 28
Models of OJA data integration 28
Scenario 1: Alternative data collection 29
Scenario 2: Integrated use 29
Scenario 3: Substitution of traditional data 30
Common OJA data processing phases 30
OJA Building blocks 31
OJA Services 35
ESS OJA implementation scenarios 37
Scenario 1 – ESS OJA implementation scenario: Replicated 38
Scenario 2 – ESS OJA implementation scenario: Shared 39
Occupation classification 40

References 43

List of Figures:

Figure 1: Comparison between OJAs and JVs for the 4th quarter 2018 in Switzerland 21
Figure 2: Scatter plot of OJAs common to X28 AG JVs datasets 25
Figure 3: OJA as alternative data collection 29
Figure 4: Integrated use of OJA and statistical data 29
Figure 5: OJA substitute statistical data collection 30
Figure 6: Common ESS OJA data processing phases 31
Figure 7: Overview of steps constituting the KDD process 32
Figure 8: ESS Reference architecture for processing OJA data 35
Figure 9: OJA application architecture 37
Figure 10: High-level building blocks that implement main OJA processing services 38
Figure 11: ESS OJA implementation scenario: Autonomous 38
Figure 12: ESS OJA implementation scenario: Shared 39
Figure 13: ESCO vs. ISCO - mapping the occupation level 40
Figure 14: ESCO different levels representation in CEDEFOP dataset 41
Figure 15: General steps in the process of ESCO code detection 42

List of Tables:

Table 1: Labour supply versus labour demand 10
Table 2: JVs vs OJAs 17
Table 3: Potential indicators from OJAs 18
Table 4: Attempt to extract pseudo-stocks for the 13 countries with CEDEFOP OJVs data 26
Table 5: Main OJA building blocks 32
Table 6: Main OJA Services 35

Purpose and scope

The purpose of this reference methodological framework is to describe at the conceptual level complete statistical online job adverts data (OJA) processing pipeline according to the four main phases: data ingestion, data pre-processing, information extraction and analysis, and production of statistical indicators. It addresses key conceptual, methodological, technical models and organisational principles and guidelines in using OJA data in official statistics. The idea is that such a framework would be used by statistical agencies as a reference guide when:

- Producing statistics using OJA data
- Integrating shared components developed by other statistical agencies or
- Designing common ESS OJA data processing system

The framework clarifies how OJA data can be used to go beyond the frontiers of conventional approaches of traditional job vacancies statistics (JVS) and identify other possible statistics beyond JVS, which could be produced from OJA data. Traditional statistical approaches, based essentially on surveys, have important caveats: cost, timeliness, accuracy, usage, integration and coverage. OJA data seem to have potential to supplement and enrich traditional statistical data and allow more granular, space-related insights in almost real time, as well as predictive analysis.

Adherence to common standards increases the consistency across the ESS, facilitates sharing and comparing methodologies, services, statistics and data. It also eases software maintainability and identification of risks and potential failures.. The aim of this framework is therefore to:

- Provide common language for various stakeholders
- Encourage adherence to common standards, specifications, methodologies
- Provide consistent methods for implementation
- Illustrate various implementation scenarios by identifying common components, processes and systems
- Provide a reference for statistical agencies and other parties to understand, discuss and compare OJA processing systems

Methodological considerations of this framework take into account the developments and conclusions of successfully concluded ESSnet Big data I [1]. Along with the production of relevant methodologies, recommendations and specifications, it was stated that current methods for producing robust job vacancies statistics based exclusively on OJA data were not sufficiently mature [2]. OJA data most likely cannot replace existing job vacancy statistics required by EU regulation. On the other hand, OJA data can provide more granular insights than traditional estimates usually offer. The role of OJA data within official statistics is more likely to provide a basis for producing supplementary indicators.

As far as technical architecture is concerned, this framework proposes the development of a convergence layers system that decouples the collection and pre-processing of alternative data from the statistical concepts, as advised by [8].This framework also discuss the role of potential common ESS OJA data collection and processing system as an implementation solution of layered concept (for the time being, based on the OJA system developed by CEDEFOP[1]). Such an approach enables NSIs to

[1] European Centre for the Development of Vocational Training

focus on remaining statistical challenges and further methodological development. This document introduces general concepts of a layered system and defines roles and responsibilities of the main stakeholders. Various scenarios for data governance, management and integration – all of them being compliant with statistical regulatory requirements - are proposed.

The target readers and users are statisticians, researchers, developers, methodologists developing OJA production systems and statistical indicators in the frame of online job adverts data.

Methodological considerations around OJAs

The rapid technology development in recent decades and digitalization of all life areas, the automatization of a number of processes in production and services, as well as the use of artificial intelligence and the growing needs of the consumers are the main prerequisites for the acceleration of the process of emerging of new professions and significant changes in the existing ones. These processes also predetermine the necessity of new knowledge and skills of the labour force. After the world economic crisis of 2008-2011, the national economies of the European Union countries started growing, which in turn led to growth in labour force demand. At the moment there is lack of labour force in many of the countries, especially in professions that require a high level of technological skills. This situation is also aggravated by the processes of population aging and the migration of highly educated young people. The development of competitive and innovative European economy is based on the need for qualified labour force with specific knowledge and skills. In this regard in June 2016 the European Commission adopted a new agenda for development of skills – New Skills, related to the above stated political priority – a new boost for jobs, growth and investments. The goal of the agenda is to ensure that the right training, skills and support will be available to the EU citizens, so that they could be equipped for quality jobs and realize their potential of confident and active citizens. This will raise the aptitude for employment, competitiveness and growth in Europe (https://ec.europa.eu/social/main.jsp?catId=1223).

The up-to-date and timely statistical information on the availability of job vacancies and on the requirements for specific skills for taking the jobs is a key element in preparing adequate policies for employment, education, including continuing education, retraining and career development at European, national, regional and personal levels. For almost two decades the job demand and supply has been carried out via the Internet. Daily job vacancies ads are published on specialized portals, companies' websites, social networks and mobile apps, and potential workers and employees submit their CVs. The huge volume of information for the demand and supply of labour on the Internet generates the so-called big data, which together with data of traditional surveys could be used for statistical study of this phenomenon and providing real-time data to the labour market.

The economic uplift in recent years, which is a characteristic feature for most of the EU countries, led to unemployment reduction. This in turn reflected in the rise of demand and the use of all sorts of recruitment channels. In active use are specialized web portals, ads on the companies' websites, social media and recruitment agencies. Using data of these online ads has a number of advantages. Furthermore, the growing computer literacy of the population predetermines the Internet as the main channel for searching and finding the right workplace. The big data for the job vacancies

generated on the Internet could be used successfully for the needs of official statistics for complementing traditional sources for job vacancies statistics. Their main advantages are their relevance, timeliness, comparatively low costs for their producing and lack of or minimal burden for the respondents. Being unstructured, most online job ads contain information not only for the vacant position but also for the required qualifications and skills, including the so-called soft skills, for degree and kind of the graduated education and professional experience, necessary for the potential applicants for taking it. Often they also contain information on the employer (sector of the economy, size of the enterprise, kind of ownership, etc.). By webscraping of on-line job vacancies (OJAs) estimates could be made for labour demand by economy sectors, professions, specific skills and qualifications, and regions. A comparison could also be made for which job vacancies similar education, qualification and specific skills are required.

However, using big data as an alternative information source for statistical surveys in any case is accompanied with a lot of challenges. In general, they are related to the impossibility or uncertainty in defining the statistical unit, with the incomplete coverage of the target population and impossibility to evaluate what part of the target population is covered, as well as to what extent certain subpopulations are covered more than others. (Giulio Barcaroli, G. at all, 2016). When using big data for studying the job vacancies the challenges are specifically related to the following problems:

- Defining the scope of the study. The OJAs refer only to part of the actual job vacancies and not all workplaces are announced online regardless of the kind of channels. Some positions are more likely to be published on the Internet and the received estimates will contain deviations connected with the occupations and qualification. As a rule, workplaces are announced on the Internet to a much greater extent in the high-tech economy sectors, as well as in such requiring higher education degree and specific qualification.

- There is not a certain information source for job vacancies and sometimes they are published simultaneously on different Internet channels – on specialized websites for publishing job vacancies ads, on recruitment agencies' websites, enterprises' websites, on social media, etc. Often some websites republish ads from other ones. This raises the issue of constructing methods for avoiding the so-called duplication of one and the same ad in one and the same Internet source or in different Internet sources and solving of this same issue is among the main challenges for using big data for monitoring job vacancies.

-

- OJAs often contain both structured and unstructured elements. In the texts of the ads in unstructured form are the usually required education, qualification and skills, which should be

turned into structured elements through special procedures. This requires using of specific classification algorithms adjusted for the respective language, which is used by the published ad. However, the data received through these algorithms contain in any case inaccuracies and errors.

- In the various ads the information about the vacancies is with different extent of details. Most often, the ads of the recruitment agencies do not contain employer data. This leads to some difficulties of collecting data for the industry sector, as well as of directly connecting OJAs data to the official statistical data. Although it's able to learn the industry sector about the on-line job position through machine learning, the classification errors could occur when processing unstructured data.

All these challenges necessitate careful searching for ways and approaches for estimating the job vacancies via information from online ads.

-

All these challenges necessitate careful searching for ways and approaches for estimating the job vacancies via information from online ads.

This document is aimed at revealing the opportunities for using the big data in statistical surveys on the job vacancies in the EU, as well as highlighting the advantages and challenges of this approach for collecting data for complementing the information collected with the traditional statistical methods in this area.

Labour market main concepts

The labour market is usually described and seen in terms of balance between supply and demand in labour force. In public statistics, data on labour supply come mainly from the Labour Force Survey (LFS) approach, while the need of data on demand is fulfilled by surveys conducted in the universe of business or enterprises. Although the Job Vacancy Statistics (JVS) are included in the Eurostat LAMAS working groups, the JVS belong from a practical point of view to the realm of business statistics. Table 1 summarizes the main concepts of both approaches of the labour market, supply and demand in labour force.

Table 1: Labour supply versus labour demand

	Definition	Main sources
Labour supply (LS)	Workforce supplied by the physical persons to the economy LS= Employed+unemployed persons (according to ILO concepts) Coverage: NACE 1-99 Threshold: 1 hour worked/week	Population registers or labour force surveys
Labour demand (LD)	Workforce the economy demands to the physical persons Inland concept; Coverage: NACE 5-98 LD=occupied posts + vacant posts Threshold: variable (from 1 to 10 employees) Vacancy rate= vacant posts / LD	Business registers or business surveys

Job adverts main concepts used

As discussed in the previous work by the Big Data ESSnet WP1[2], there are fundamental methodological issues around the quality of online job advertisements (OJAs). These issues need to be analysed in order to use these data to produce supplementary indicators for official statistics In particular, the main ones regard what the OJAs represent and cover and how they can be compared with official estimates produced from the official job vacancy survey..

The job vacancy advertisements on job portals (OJAs) cannot be considered similar to the job vacancies (JVs) as defined in the EU Regulations[3]. There are differences between the measures which derive from the definition used by the European official surveys and what can be practically measured from on-line job advertisements.

In particular, official job vacancy statistics, regularly transmitted to Eurostat by Member States and also published in the Eurostat online datawarehouse, are subject to three different EC Regulations[4].

Official job vacancy statistics provide information on unmet labour demand. In particular, quarterly data on job vacancies are used by the Commission (the Directorate-General for Employment, Social Affairs and Inclusion and the Directorate-General for Economic and Financial Affairs) and the European Central Bank to monitor short-term developments in the business cycle and the labour

[2] Web Scraping for Job Vacancy Statistics in the SGA-2 (that ran from August 2017 to the end of May 2018) and in the SGA-1 (February 2016 to July 2017).

[3]Job vacancy statistics (JVS) are collected under the framework regulation and two implementing regulations: the Regulation (EC) No 453/2008 of the European Parliament and of the Council of the 23 April 2008 [11] on quarterly statistics on Community job vacancies; the Commission Regulation (EC) No 1062/2008 of 28 October 2008 implementing Regulation (EC) n. 453/2008 of the European Parliament and of the Council on quarterly statistics on Community job vacancies, as regards seasonal adjustment procedures and quality reports; and the Commission Regulation (EC) No 19/2009 of 13 January 2009 Implementing Regulation (EC) n. 453/2008 of the European Parliament and of the Council on quarterly statistics on Community job vacancies, as regards the definition of a job vacancy, the reference dates for data collection, data transmission specifications and feasibility studies.

market. The job vacancy rate is one of the principal European economic indicators (PEEIs), the primary source of information used to analyse and monitor short-term cyclical economic developments within the EU, the euro area and individual EU countries. JVs are also used for indicator-based structural analyses conducted as part of the Europe 2020 strategy for smart, sustainable and inclusive growth. The job vacancy rate is also included in the employment performance monitor, a selection of key indicators in the labour market that are monitored in the context of the Commission's Joint Employment Report (JER).

On the basis of the EC Regulation No 453/2008 a job vacancy is defined as:

"*a paid post that is newly created, unoccupied, or about to become vacant:*

a) for which the employer is taking active steps and is prepared to take further steps to find a suitable candidate from outside the enterprise concerned; and

(b) which the employer intends to fill either immediately or within a specific period of time."

The active search for a suitable candidate must have already started but not yet ended at the reference time at which job vacancies are measured. The active steps to find a suitable candidate include: (i) notifying the job vacancy to the public employment services; (ii) contacting a private employment agency/head hunters; (iii) advertising the vacancy in the media (for example the Internet, newspapers, magazines); (iv) advertising the vacancy on a public notice board; (v) approaching, interviewing or selecting possible candidates/potential recruits directly; (vi) approaching employees and/or personal contacts; (vii) using internships.

EC regulation 453/2008 has several mandatory elements:
• Quarterly data that have been seasonally adjusted
• Data broken down economic activity (using NACE)
• Data are relevant and complete, accurate and comprehensive, timely, coherent, comparable, and readily accessible to users

There are other elements that have been optional until now, or subject to feasibility, including:
• Job vacancies in the agriculture, forestry and fishing sectors
• Job vacancies in public administration, defence and education
• Data on businesses with fewer than 10 employees
• Distinguishing between fixed term and permanent jobs

However, the task force that has been currently working on the review of the legislation in the field of Labour Market Indicators (LMI) will define as mandatory also the extension of JVs data to the public sector and small enterprises with fewer than 10 employees.

The just described definition of a job vacancy in the EU regulation framework makes it clear that the concept of OJAs does not correspond to that of JVs, due to different factors.

In particular, the online job vacancies should cover only those job vacancies, as defined by the EU Regulation, for which the active steps carried out by the employers to find a suitable candidate include also the advertising on Internet job portals. But, not all job vacancies are advertised on-line:

although there is a general trend towards more job vacancies being advertised on-line, many continue to be filled through traditional channels, such as newspapers, employment agencies, noticeboards, or personal contacts. The results of the Slovenian study within the WP1 pilot suggest that only about 40% of all job vacancies in Slovenia are advertised on-line. A similar proportion has been reported for France, although this figure may be higher in other countries.

Furthermore, job portals used by the employers are not necessarily totally covered by the web scraping and other data ingestion activities. This is due to the fact that there is no definitive source of OJAs. Even if the situation varies between countries, OJAs data are generally characterized by multiple job portals, with different business models and complex interplay between them. Job boards only publish original ads uploaded by employers. Job search engines republish ads from other portals. Hybrid job portals are a combination of both. Some portals advertise many different types of jobs while others may specialize in specific sectors. In addition, new job portals may appear while existing portals may decline in importance. All these factors mean that it is very difficult to capture all jobs advertised on-line.

However, even if all the online portals used for the job advertisements were considered, it is not clear if all the job advertisements posted on all job portals refer to total job vacancies that actually exist at a specific reference date. A job ad is only a proxy measure for the existence of a job vacancy within a company.

For example, delay in the communication between enterprises and the online portals could cause the presence on portals of job vacancies that are closed (because, for example, a suitable candidate has already been found). This should mean that the stock of vacancies as measured by the OJAs should be higher than the corresponding JVs data as measured by the current official surveys on job vacancies. In addition to this, either some OJAs may not represent a job vacancy in the scope of the official survey (these include non-existent vacancies, referred to as “ghost vacancies”, international jobs, and non-paying student internships) or some OJAs may contain more than one vacancy.

Furthermore, the number of vacancies collected by the official job vacancies survey is the stock of vacancies for which businesses are actively seeking recruits from outside their organization at a specific reference date. Online job ads represent a flow of new vacancies but usually do not contain explicit information on when the recruitment activity started and when it will be concluded.

The above mentioned consideration contributes to making a different representativeness of the OJAs with respect to the JVs as defined by EU regulations.

Online job vacancies: seen as proxies to statistical indicators

Like job advertisements in newspapers before the introduction of the formal Eurostat JVS regulation, OJAs can be seen as proxy:

- To explain the link between JVS and OJA
- If our proxy is sufficiently congruent with JVS, we can safely use OJA

In statistics, a proxy or proxy variable is a variable that is not in itself directly relevant, but that serves in place of an unobservable or immeasurable variable. [10] (Wikipedia https://en.wikipedia.org/wiki/Proxy_(statistics)#cite_note-1). A proxy is an indirect measure of the

desired outcome which is itself strongly correlated to that outcome. It is commonly used when direct measures of the outcome are unobservable and/or unavailable. An organization should use a proxy measure when there is little or no data available about the program being implemented, but the outcome the program is designed to influence has an existing and commonly accepted proxy.

Modelling is the way to use proxies in order to produce statistical indicators. Regression is a simple tool to identify correlations and produce reliable predictors to statistical indicators.

Short description of the JV survey process

EU job vacancy statistics are compiled on the basis of data provided by the participating countries. Member States are granted considerable flexibility regarding the implementation of Regulation 453/2008 in the national statistical systems. Some countries use stand-alone surveys, while others combine the job vacancy survey with other business surveys. Although the Regulation states that the data shall be collected using business surveys, the use of administrative data is equally permitted under the condition that the data are "appropriate in terms of quality" (according to the quality criteria of the European Statistical System). All the countries, except Czechia, Luxembourg and Croatia, conduct a survey to estimate the number of job vacancies. Czechia, Luxembourg and Croatia use administrative sources.

The minimum information required is the number of job vacancies and the number of occupied posts. Data on occupied posts are generally obtained from the same survey as data on job vacancies.

Surveys are conducted mainly by using postal questionnaires or Computer Assisted Telephone Interviewing (CATI) or Computer Assisted Web Interviewing (CAWI).. As mentioned, administrative sources are also used.

The basic statistical unit for the data collection is the establishment or local unit. However, the majority of participating national statistical offices collect the vacancy data from enterprises.

As regards, the statistical population, the data cover all enterprises with one or more employees. In France, Italy and Malta only business units with 10 employees or more are surveyed.

Data on job vacancies and occupied posts represent the stock of vacant and occupied posts at a given reference date or averaged across a given reference period. There is currently no internationally agreed rule for the time of recording of job vacancy statistics. Depending on countries, the time of recording for quarterly job vacancy statistics may be one specific day in the quarter (e.g. the 15th of the middle month, the last calendar or working day of the quarter) or a three months average.

As required by the framework regulation, Member States are required to provide the number of job vacancies and the number of occupied posts that can be considered representative for the reference quarter. The preferred methods to achieve this are data collection on a continuous basis or the calculation of a representative average of the data collected for specific reference dates.

The data cover all economic activities defined by NACE Rev. 2, the common classification system for economic activities, except for the activities of households as employers and the activities of extraterritorial organizations and bodies. Covering agriculture, forestry and fishing activities is optional. Denmark provides data only for NACE sections B-N. In France and Italy, in the case of public

administration, education and human health (NACE Rev. 2 sections O, P and Q), public institutions are not covered.

The response rates of the surveys vary between 62% and 99% amongst the different Member States.

The quarterly data are broken down by economic activity (at section level), in accordance with NACE Rev. 2 (Statistical Classification of Economic Activities in the European Community). In addition, Eurostat publishes for some countries the quarterly data broken down by occupation (in accordance with ISCO nomenclature) and region (in accordance with NUTS nomenclature), which are delivered by the countries on a voluntary basis.

Data are required to be seasonally adjusted. It is mandatory to supply seasonally adjusted data on job vacancies and occupied posts at least for individual/aggregated NACE Rev. 2 sections B-E, F, G-I, J, K, L, M-N, O-Q, R-S. Eurostat uses the data supplied on seasonally adjusted job vacancies and occupied posts to calculate the seasonally adjusted job vacancy rates for each country as a ratio.

Member States whose number of employees represents more than 3% of the European Union total transmit the aggregate number of vacancies and occupied posts within 45 days after the end of the reference quarter. Eurostat publishes JVS flash for the EU around 5 days after the deadline for the data transmission. Within 70 days after the end of the reference quarter, Member States transmit data broken down by economic activity. Eurostat publishes a news release around 8 days after the deadline for the data transmission.

Peculiarities and weaknesses of Job Vacancy statistics

Job vacancy statistics have been discussed and described in great detail during two Eurostat workshops whose major results are summarized in [12]. They present some unique peculiarities, as well as weaknesses or unsolved questions which are described below

a) **Most units report 0 vacancies:** job vacancies are a very peculiar case among statistical variables in the sense that in surveys most responding units report 0 vacancies, while most vacancies are reported by a few units, mostly by the largest enterprises. For instance, in the Swiss JVS survey for the 2nd quarter 2008, 40’500 of the 47’700 responding units reported zero job vacancies (85%). This share was highest among smaller units (96%), with only 4% of the smallest firms reporting at least one vacancy (class 1 : <5 FTE (FTE=full time job equivalents)), 85% among class 2 ([5-20[FTE), 69% among class 3 ([20-50[FTE) and 47% among the largest local units (class 4 >50 FTE).
b) **Business registers do not record information on job vacancies:** job vacancies do not belong to the standard list of variables of business registers. As a consequence, JV survey designs are therefore based on other kinf of information as auxiliary variables for sample drawing and grossing up (e.g. number of jobs, employees, VAT, etc.).
c) **Lack of auxiliary information for the treatment of missing values**: because of the absence of auxiliary information relevant to job vacancies, the treatment of missing values in JV surveys is still an open and often unsolved question.
d) **Job vacancies in the branch “Employment activities” (NACE 78):** jov vacancies in the branch “Employment activities” is still an open and debated question. Reliable data on Job vacancies are very difficult to get in this branch and their distribution, either in the NACE 78 or in the paying sector are still under discussion in the LAMAS group.

OJA web scraping in the context of the big data life cycle

The big data processing lifecycle for the production of official statistics (ESSnet on BD I, IT Report 2018) is almost similar to the Generic Statistical Business Process Model (GSBPM 5.1, 2018). In general, a big data process is considered to be composed of 5 main phases:

Acquisition and recording

Extraction, cleaning and annotation

Integration, aggregation and representation

Modelling and interpretation

Dissemination

The above-mentioned big data life cycle phases are identified as the Collect, Process, Analyse and Disseminate phases in the GSBPM 5.1, respectively.

In the first step *Acquisition and recording,* it is all about getting access to the OJAs through web-scraping technologies to acquire and store OJAs data. Typical for this step is that the data are made available in their 'raw' (unprocessed) form as html text files and need to be processed in the next phase in order to be ready for further analysis.

The process of surveying of the job vacancies from the information published on the Internet begins with a choice of sources. Online portals, Internet websites of companies and social networks could be pointed out as main sources. The online portals are public and private. The public portals are most often created, maintained and managed by public employment agencies at national, regional or local levels. The private portals could be owned and managed by national companies or by international consortia. There are also portals which collect ads published elsewhere. Some recruitment agencies also have their own Internet portals. Besides, many companies, especially the bigger ones, have specialized sections for job vacancies ads on their own Internet portals. There are also such sections on some online media, as well as on portals of any kind. In the social networks all these portals could have their own pages, as well as there could exist specialized pages and groups for publishing job vacancies ads. There exist also search engines of job vacancies crawling the portals and companies with published ads by pre-set criteria. In recent years the so-called hybrid portals became popular. They combine their own portal where the employers could publish ads with a search engine in other portals. Some of them offer the employer's standard interface by which the ad uploaded on the company website is accessible also via the portal.

The next two steps *Extraction, cleaning and annotation and Integration, aggregation and representation* cover everything necessary for processing and preparing the raw scraped OJAs data prior to analysis. The major concern here is the quality improvement of unstructured OJAs data.

Because of that techniques and methods for data cleaning, deduplication, data classification, data adjustments, etc., editing sub-processes are applied.

After the stage of collecting OJAs data it is necessary to carry out preliminary processing of the received information aimed at structuring in the form of a database with statistical information. All sources contain a text for the job vacancies which could not be processed through traditional statistical methods. That is why via different techniques for processing of the text content it should be structured in a proper way for extracting the meaningful statistical insights. At this stage it is necessary to remove the so-called noise, which represents information that is not related to the job vacancies, e.g. ads, presentations of companies and opportunities in drop down menus. If this noise is not removed, the information would be incomprehensible and improper for the aims of the statistical analysis. Also it is of key importance to avoid the repeated counting of job advertisements between or/and within some portals. As mentioned, often one and the same job vacancy is published on several different places. So it is necessary that the information be collected and based on pre-set criteria the ads for one and the same workplace to be identified and the duplications be removed.

Part of the preparation may be combining the cleaned OJAs data with administrative sources, e.g. job vacancies data from the Public Employment Agencies at national level (if they are available). Although the OJAs data may still be unstructured at this stage, as a consequence of the adjustments made, the data become more and more structured. This process is not linear and contains many cycles as findings at the end of the chain initiate the need to improve some, or a number, of the previous steps.

The *Modelling and interpretation* step is referred to analysing of processed OJAs data from the previous step by applying the machine learning based approaches, using Python, R, etc. The final goal of this stage is to produce new insights and then to calculate the statistical indicators for OJAs. These indicators can be used as a proxy to measure the job vacancy market at national and EU levels. In the last step *Dissemination* it is all about producing statistical outputs and then visualizing them to make the data better understandable for analysts.

Possible statistical indicators

"….OJV data cannot be used to directly replace the existing job vacancy statistics that are required by EU regulation. Indeed, the quality issues are such that it is not clear if these data could be integrated in a way that would enable them to meet the standards expected of official statistics. On the other hand, OJV data can provide more granular insights that official estimates usually offer. It was concluded that the role of OJV data within official statistics is more likely to be as the basis for

producing supplementary indicators." (ESSnet Big Data Specific Grant Agreement No 2 -SGA2, Final Technical Report, version 2018-05-30, p. 5).

The official data on job vacancies, as required by the EU Regulations and regularly transmitted to Eurostat by Member States, are collected primarily for the purposes of calculating the job vacancy rate, which is one of the Principal European Economic Indicators (PEEIs) for the labour market monitoring.

The job vacancy rate measures the proportion of total posts that are vacant, according to the definition of job vacancy above, expressed as a percentage as follows:

$$\text{Job Vacancy Rate} = \left(\frac{\text{Number of job vacancies}}{\text{Number of occupied posts} + \text{Number of job vacancies}} \right) \times 100$$

The job vacancy rate is required at national level, disaggregated by economic activity sections, enterprise size and on a quarterly basis.

The official job vacancy statistics are not required by EU Regulation broken down by other characteristics of the vacant job position, such as the geographical area/region where the vacant post is located and the profession required. However, these kinds of characteristics are considered as particularly useful by current and actual users, and even if the production of job vacancy indicators broken down by these two characteristics is not required as obligatory by the current EU Regulations, Eurostat already invites Member States to disseminate and transmit these data, if available. For some Member States these data are already available in the Eurostat online data warehouse. Furthermore, the possibility to make obligatory the production of job vacancy statistics broken down by region and profession once a year, in one of the survey waves, is going to be discussed by the Task-force on the Review of EU legislation in the field of Business Based Labour Market Indicators (LMI), whose works are currently in progress.

Table 2: JVs vs OJAs

Dimension	**Official JVs**	**OJAs**
Timeliness	70 days after the reference quarter and 45 days[5]	Real time
Frequency	Quarterly	Daily
Economic activity sector	Yes	Yes
Enterprise size	Yes	Yes
Occupation	No	Yes
Skill	No	Yes
Education	No	Yes

[5] Only for Member States whose number of employees represents more than 3% of the European Union total.

Geographical area/region	No	Yes
National totals (estimate)	Yes	No

Moreover, additional information on a monthly basis and within the month at daily level, as that deriving from the OJAs, gives useful insights into the dynamics within the quarter of the labour demand expressed by the online recruitment process. . In general, despite the representativeness limits of OJAs, daily information on them provides valuable insights into the vacancy flows during the quarter and not only into the vacancy stocks at a specific date of the quarter. In the current official vacancy surveys job vacancy data are referred to one specific day in the quarter (e.g. the 15th of the middle month, the last calendar or working day of the quarter) or to a representative quarterly average. Therefore, daily OJAs allow observing the total flow of vacancies opened and closed within the quarter and not only those vacancies that survived at a specific date of the quarter.

Furthermore, the breakdown by economic activity, occupation, region, education together with data at a frequency higher than the quarter could provide useful insights into the change of the job positions characteristics that need to be filled by the enterprises.

Amongst the concluding remarks of the previous phases of the project, it had been shown that while "…it *is feasible to produce estimates of on-line job vacancies, there is a big difference between these and the official job vacancy estimates. It is therefore clear that these could never replace the official estimates. Further, the benefits of these estimates for policy making are not clear as they only give a partial (and not easily defined) view of overall labour market demand. It therefore seems that the role of OJV data within official statistics is more likely to be as the basis for producing supplementary indicators. These could include indicators of local labour market demand (as shown above) and/or indicators of occupation groups and associated skills. However, rather than measuring absolute levels, these would be more useful for measuring change over time*…"(ESSnet Big Data Specific Grant Agreement No 2 -SGA2, Final Technical Report, version 2018-05-30, p. 35)

Therefore, detailed *supplementary* indicators could be derived from OJAs.

Table 3: Potential indicators from OJAs

Indicator	**Frequency**	**Grouped by**
Number of available on-line job vacancies in a reference day	daily	economic activity, occupation, region, education
Number of available on-line job vacancies in a reference month	monthly	economic activity, occupation, region, education
Number of newly posted on-line job vacancies in a reference day	daily	economic activity, occupation, region, education
Number of newly posted on-line job vacancies in a reference	monthly	economic activity, occupation, region, education

month		
Average number of OJAs	weekly , monthly, quarterly	economic activity, occupation, region, education
Percentage changes over the same period of the previous year (year on year changes) of the number of OJAs	daily, weekly, monthly,	economic activity, occupation, region, education
Percentage changes over the same period of the previous quarter (quarter on quarter changes) of the number of OJAs	daily, weekly, monthly,	economic activity, occupation, region, education
Number of available on-line job vacancies at the JVs reference day	quarterly	economic activity, occupation
Daily pseudo-stocks of job ads based on CEDEFOP data	quarterly	industry sector, region or required qualification

Complementation of JVS

OJAs can be used in several ways to complement the official JVS, in particular as a complementary source :

a) as a new set of variables in business registers. Once JVs included in business registers, it would be possible to design surveys more specifically tuned to job vacancies, taking into account the actual distribution of JVs among enterprise sizes and possibly to reduce the burden on enterprises. This would of course imply a full coverage program of all enterprises

b) to fill non-response units and for imputation purposes in general

c) as auxiliary variables during JVS grossing up processes

d) to improve the JVs data collection of NACE 78

The choice of websites for recording JVs and the quality of the linkage between OJAs and register units are a key issues in this context.

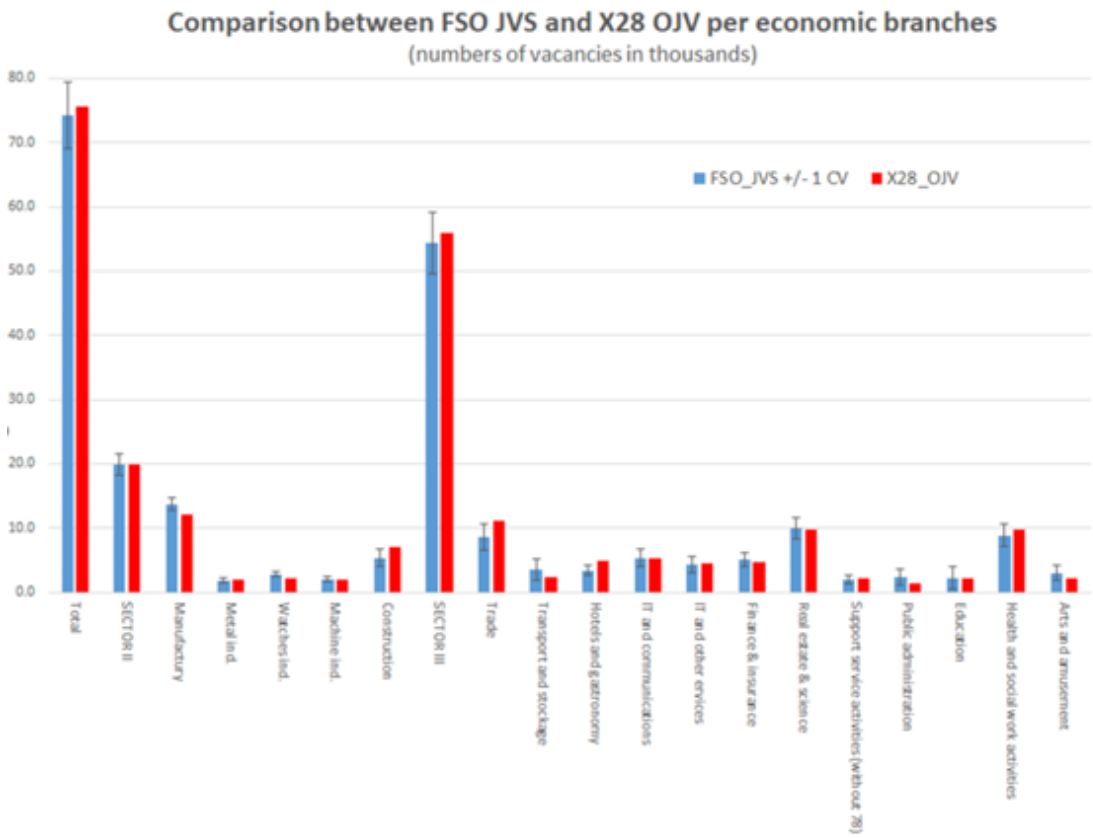
Choice of websites: enterprise websites versus job portals

Until now, most attention has been turned towards job advertisements published by major job portals, which are partly a secondary source of data, since they tend to collect and republish job advertisements from enterprise websites or from other portals. An alternative to this approach would be to collect primary source data, i.e. job vacancies published by enterprises on their own websites.

A good example of such an approach has been conducted for Switzerland comparing data collected by the private company X28 AG with the official JV survey [13]. The analysis of the data for the 4th quarter 2018 shows that a full coverage of primary source data is a realistic alternative to survey data (Figure 1). Primary source data, i.e. vacancies collected online from websites enterprises, have been matched to enterprises of the business register. Using the UID as a matching key (https://www.bfs.admin.ch/bfs/en/home/registers/enterprise-register/enterprise-identification/uid-entities.html), a unique match could be found for most units, i.e. 23'600 of 25'200 enterprises, i.e. 93,7%. The remaining 6,3% unmatched units encompassed 3'700 vacancies, i.e. 4.4% of the 85'100 job vacancies of the the X28 AG test file. Filtering the matched data for Sections B to S of the NACE and excluding NACE 78 which is not covered by the Swiss JVs statistics resulted in a total amount of 75'600 vacancies from online websites with reference data Dec. 15th 2018 as compared to 74'300 ± 5'100 vacancies for the official JV survey with reference date Dec. 31th 2018. Similar results were found at the regional level, with OJAs within the confidence intervals of the JV survey for the seven NUTS2 regions.

More interestingly, a full coverage of all enterprises of the country (ca 600'000) resulted in a OJA dataset of a ca 25'000 enterprises with at least one job vacancy (4.2%), a size similar to the JVs survey (22'000 enterprises; 3.7%).

On the other hand, the total number of online job vacancies recorded by X28 AG, i.e. including vacancies recorded on job portals, and published in the press release Jobradar for the same quarter amounted to 189'900 (https://www.jobagent.ch/resources/jobagent/jobradar_Q4_2018_de.pdf), which indicates that the official number of job vacancies represents only 40% of the total number of advertisements collected online, or a ratio OJAs/JVs of 2.5. This ratio has been increasing since 2013, with an average over the last seven years of 2.4 with a minimum of 1.8 in 1st quarter of 2013 and a maximum of 2.9 in the 4th quarter of 2016.



Legend

Blue bars: JV survey estimates with 95% confidence intervals; red bars: OJAs collected from enterprises websites (primary source data). The first pair of bars from the left corresponds to the numbers of job vacancies for whole Switzerland, the second pair to "Industry & construction: sections B to F," and the eight pair to "Services: section G to S (NACE 78 excluded)".

Figure 1: Comparison between OJAs and JVs for the 4th quarter 2018 in Switzerland

Some quality aspects of OJA data

According to the work done by ESSNet Big Data I, WP8 Methodology, there are five main groups of measures that can be used to assess the data quality in the web scraping process of online job vacancy data: coverage, accuracy and selectivity; measurement errors; comparability over time; linkability and processing errors; model errors and precision. In the current ESSnet Big Data II project, the Quality Guidelines for the Usage of Big Data is being prepared by the WP-K Methodology, where the web scraping is described as a separate big data class.

Coverage and selectivity

As previously discussed, one of the fundamental quality issues in using on-line job advertisement data is that not all job vacancies are advertised on-line.

Understanding these issues of coverage has been a key focus of pilot studies carried out in the previous phases of the project, aimed at identifying different approaches for trying to better understand and measure these differences (SGA - 2 Final Technical Report, p 28). In particular, three different approaches had been defined: micro-level comparisons, aggregate comparisons and measuring use of advertising channels via the JVS. The first approach "...involves linking OJAs data to either the reporting units of the JVS or the Business Register. The UK has linked JVS data to vacancy counts by company for a range of OJA sources. This has revealed that the pattern of job counts between the JVS and other sources by reporting unit/company is typically very messy and difficult to understand…". The second one "...involves comparing JVS aggregates by industry sector (i.e. NACE) with the equivalent taxonomies from job portals. This approach has been used by Germany, who does not have access to JVS micro data. While this is quite a straightforward approach, private portals usually have their own taxonomies, which are often only approximately comparable with NACE…". A similar attempt recently conducted for Switzerland proved to be a very promising approach when a good linkage between OJAs and enterprises can be achieved, with OJAs close to JVS estimates at the economic activity section and regional levels. A third approach "…has involved surveying enterprises and asking specific questions about advertising channels. Surveys of this type have been carried out by both Germany and Slovenia. The German study found that large companies were more likely to advertise on-line, with small companies more likely to use other channels." (SGA - 2 Final Technical Report, p. 28). All these three approaches can be use together in order to better understand the actual coverage differences between OJAs and JVs. By means of these approaches over and under coverage issues can be deepened throughout the analysis at the enterprises characteristic level: enterprise size, economic activity sector and geographical area.

As other big data sources, OJAs also suffer from the problem of there not being an appropriate frame for the target population, i.e. there is no list that fully covers the target population. This makes it difficult to define the size of the under-coverage.

The lack of frame for the target population does not allow calculating the known totals used in the calibration procedure. Therefore, calibrated weights cannot be computed in order to obtain OJA total estimates. As a consequence, OJAs cannot be considered as representative of the total number of vacancies that exist at a specific date.

Furthermore, in big data sources in general, which are based on non-probability samples by definition, the selection bias becomes dominant. In the field of big data sources, the discussion is restricted to biases occurring due to the self-selection mechanism, that is caused by individual sample units, and selectivity is defined as a: "*…general term for self-selection error which results from individuals (unit-specific, e.g. whether to tweet or use a certain mobile phone network provider) and data holder decisions (data holder specific, e.g. in terms of business concept, technical infrastructure). This includes coverage, measurement or non-response (missingness) error, which further introduce potential bias in estimates based on big data sources only*." (see Eurostat, 2018, p. 25)

In particular, the data collected by the web scraping activities suffer from unit-specific selectivity (self-selection) - that is the result of any decision of individuals (or other entities) that affects its presence in the sample (see Eurostat, 2018, p. 25).

In the case of big data in general, there are three types of decisions to take into account:

a) The decision to use the technological platform from which big data are being captured, e.g. the Internet or a device connected to the mobile phone network;

b) The decision to select a particular Internet platform (e.g. one particular social media portal) or a particular mobile operator; and

c) The decision to provide paradata that might be used to impute (pseudo-)responses or auxiliary variables.

In the case of OJAs, in particular, different types of enterprises can use in different ways the advertising on job portals. For example, very small enterprises may prefer to make use of more informal active steps, such as word of mouth. In addition to this, employers could consider that the advertising on job portals could be more effective for some kinds of job positions, with reference to their characteristics, such as kind of profession required, geographical area, level of education required, etc. Thus, some types of jobs might more likely be advertised on-line than others. This means that OJAs data might not only miss many jobs, but might also not be representative of the overall job market.

Measurement errors

The measurement errors of web scraping of job vacancies on job portals are mainly the result of scraping errors, i.e. the scraper may construct such as to download incorrect data from the job portals. The other type of measurement error could be the errors on the job portals or employers may upload incorrect data on the job portals.

Given the nature of these data resembling a flow, some thought has to be given to network connectivity problems, which might disrupt the collection of data massively.

Comparability over time

Multiple counting of OJAs might impede comparability over time and comparability across countries. In reality, the supply and demand on the labour market have not been changed but multiple counting introduces an additional variance and false fluctuations.

Linkability

When scraping OJAs, the goal is to find the number of ads for each company through information from the Internet, so we have to link names (as they appear on the Internet) to official company information in the Statistical Business Register (SBR). At the same time scraped job titles and descriptions do not usually match the official ones and classification codes are not included, so we classify them through machine learning - classification method and then to the official names/titles.

The linking of on-line job ads with JVS micro data or SBR data is intended to understand coverage issues of scraped OJAs data. Usually, this is a problematic issue, because the employers use abbreviated names, trading names rather than the legal enterprise name, and misalignment between the company names and the JVS reporting unit.

Linking at the enterprise level between the National Employment Agencies data and the Business Register is straightforward since they both use a common enterprise numerical identifier.

A major issue in trying to match JVS reporting units to company names in job portals is that many jobs are advertised through private employment agencies and the employer is not usually identified in the advertisement or identified with.

However, the analysis of data of X28 AG in Switzerland indicates that when sufficient energy is put into the linking process very accurate figures can be produced using OJAs from enterprise websites data.

Quality of individual matching in the Swiss example

However when cases are examined individually, the match is very poor in the example described above for Switzerland, with a very wide scatter of JVs versus OJAs, while one would expect data aligned on a straight line of slope 1 (Figure 2 - horizontal axis: OJAs collected by X28 AG from enterprise websites (log scale); vertical axis: JVs collected in the survey (log scale); N =3511 units common to the two datasets). How can this be explained? There at least two major cases :

a) the data collected online and the survey differ widely
b) the matching is incorrect

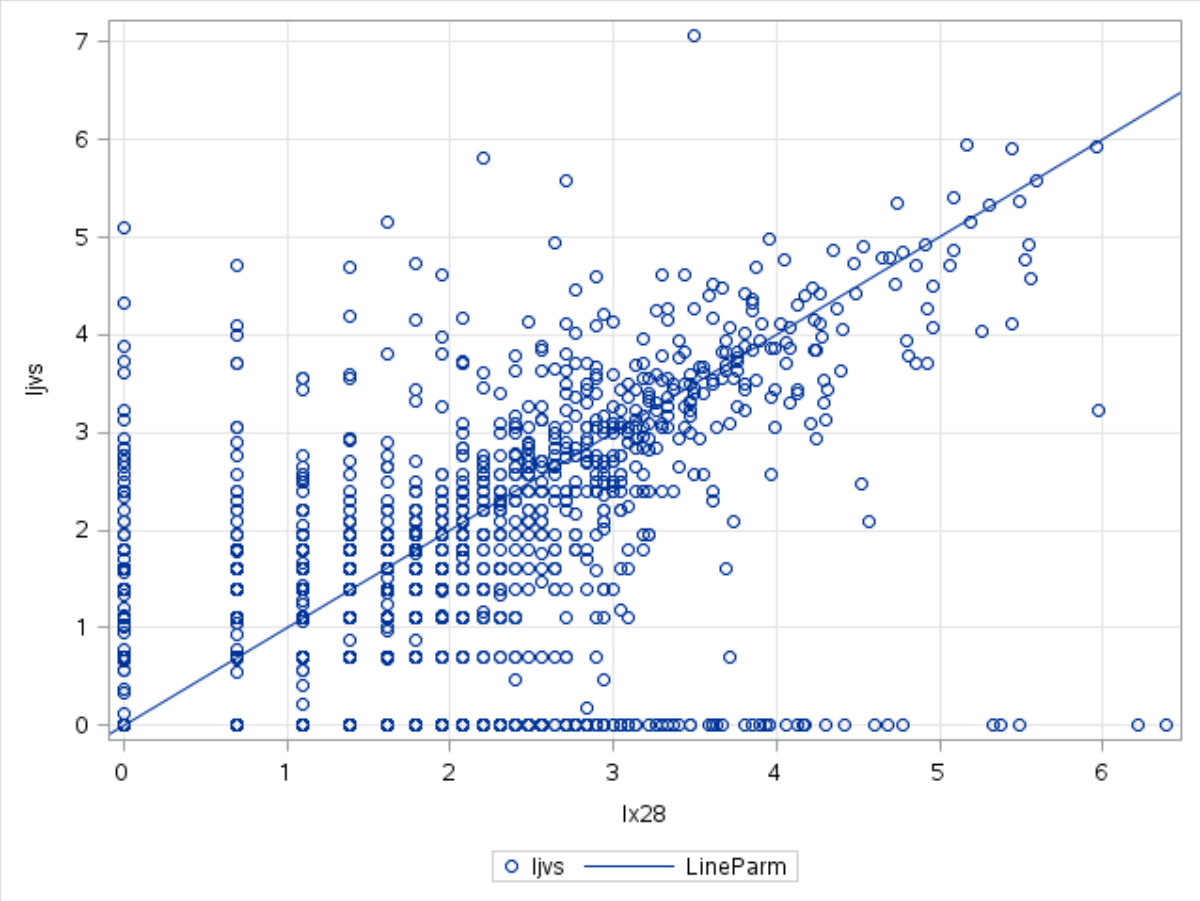


Figure 2: Scatter plot of OJAs common to X28 AG JVs datasets

A close examination of the data indicates that the matching is problematic, because of the complexity of the enterprises and of their identifiers in business registers.

Identifying enterprises in business registers

The structure of enterprises in business registers may reach a very high complexity, especially in large enterprises with many establishments and local units. Four kinds of identifiers can be distinguished, each of these categories forming a particular layer:

a) legal units (bear legal attributes with name and address)
b) administrative units (used for administrative purposes with name and address)
c) physical units (bear employees, buildings with localisation, produces goods and services)
d) statistical units (level to which statistics must be produced, e.g. local or kind of activity units)

The structure of enterprises may be also be very dynamic with frequent and profound changes. In cases of closures and reopening, fusion and acquisition, splitting, even if the physical structure may be little affected (for instance a restaurant whose owner changes may keep the same name, address menu and employees), the legal and administrative units will change, as will the statistical unit. It is no more the same "enterprise". Only specialists of business registers in NSIs are able to properly match data collected from the reporting units to the correct statistical units. Without this important matching effort and knowledge of the enterprises structures, there is little hope to produce statistical products which respect JVS quality standards using OJAs.

Pseudo-stocks

Germany proposed to approach JVs using the concept of "pseudo-stocks". "The idea behind pseudo-stocks is the following: At a given point in time (T), we consider a job ad to be valid if it hasn't expired yet and was posted a maximum of 30 days before T. Expiration of a job ad is determined by the variable *expire_date*. Preliminary analysis of other sources indicates that job ads stay relevant for roughly 30 days on average (see ESSnet WP 1). The number of all job ads which are considered at time T is the pseudo-stock of active job ads at time T". If the results seem to provide pseudo-stocks relatively close to the JVs for the total for whole Germany for the three quarter analysed (3rd quarter 2018 to 1sr quarter 2019), the agreement is not so close when the data are displayed per economic sections, which might result from NACE classification errors [14].

The same idea had been previously applied to the 13 countries for which CEDEFOP data are available with slightly more complex selection criteria [15]. When the OJAs figures differed too much from the JVs, a closer agreement was searched playing with the time window. The results are shown on Table 4. They indicate rather diverging results among countries with OJAs almost 50% higher than JVs for UK and more than 80% lower for CZ. Extending the time window to 90 days for OJAs did not allow to reach the JVs level for BE, while this level was still 30% higher with a 15 days window for Ireland. There no JVs data for IT and FR. This track does not seem very promising unless new criteria are developed.

Table 4: Attempt to extract pseudo-stocks for the 13 countries with CEDEFOP OJVs data

		JVS				OJV			Time	atio/OJV_JVS				
		2018Q3	2018Q4	2019Q1		2018Q3	2018Q4	2019Q1	Window		2018Q3	2018Q4	2019Q1	average
AT	Austria	126'335	130'869	128'211		159'253	108'497	84'430	30 days		126%	83%	66%	92%
BE	Belgium	149'180	141'730	147'698		108'292	111'870	146'790	90 days		73%	79%	99%	84%
CZ	Czechia	270'588	277'441	295'904		52'482	47'536	45'221	30 days		19%	17%	15%	17%
DE	Germany	1'227'600	1'445'897	1'366'915		1'517'218	1'732'571	1'365'818	30 days		124%	120%	100%	114%
ES	Spain	124'747	123'772	131'531		112'457	135'822	135'773	30 days		90%	110%	103%	101%
FR	France	:	:	:										
IE	Ireland	19'200	17'400	19'500		20'123	24'565	27'611	15 days		105%	141%	142%	129%
IT	Italy	:	:	:										
LU	Luxembourg	7'527	7'295	7'272		8'076	8'906	8'008	60 days		107%	122%	110%	113%
NL	Netherlands	255'843	252'782	276'107		500'000	322'338	113'765	30 days		195%	128%	41%	121%
PL	Poland	156'303	138'434	141'906		255'760	108'430	96'574	30 days		164%	78%	68%	103%
SE	Sweden	103'006	103'250	142'523		93'560	107'216	166'167	30 days		91%	104%	117%	104%
UK	United Kingdom	865'000	854'000	820'000		1'643'401	1'391'481	731'965	30 days		190%	163%	89%	147%

Processing errors

When dealing with semi-structured and unstructured descriptions of OJAs, we are dealing with mostly text data, so some form of language processing (machine-learning methods) is needed. By definition this means that either data are pre-processed into features which are substitutes for the true information stored in the text or a form of artificial neural network is used. Either way, algorithm validity errors should be considered.

The raw scraped data from job advertisements usually require a lot of processing steps before they can be analysed. For example, job title fields often contain additional information, such as job location, key skills, and salary. This could generate the processing errors.

De-duplication seemed to be another major issue during the processing phase. There is need to design the specialized software tools or methods for effective removal of the duplicate records in/between sources.

The data integration errors can happen when the OJAs data integration on several different job portals is done or when the data gained from web scraping are linked with the data from the JVS.

All these reasons make OJAs not to be considered as exhaustive of the total number of job vacancies that exist at a specific reference date.

Fictional OJAs

In some cases, advertisements may be published without actual posts to fill, for example to collect CVs or for marketing or communication purposes. Such cases are virtually impossible to detect.

ESS OJA reference production architecture

The current methods in official statistics are based on specific features of traditional statistical data, small amounts of high quality data. These data are collected to fit a specific and predefined statistical purpose and are usually well structured. They are usually collected to fit a specific and predefined statistical purpose to ensure a given coverage of population, definitions, methodology and quality in order to meet the needs of official statistics. The guarantee that statistical data are of good quality is the responsibility of the domain expert who designed the survey. This scenario changes when dealing with OJA data. These data can have any kind of structure, they may be structured or semi-structured (data collected from job portals) or completely unstructured (companies' web pages). These data are scraped continuously from the Internet, with dynamic changes of webpages and job portals. The quality of data depends on ability in collecting and pre-processing the data (identifying duplicated adverts, etc.) and has to be linked to registers, as well as combining with other statistical data sources. These differences between OJA data as opposed to JVS data also force to shift the approach to the analysis: from a model-based approach using a top-down process to a data-driven approach requiring a bottom-up method.

There is a need for developing new models and tools to support the integration of alternative data sources into official labour statistics. As described in [7] the focus should move to the production of multi-source statistics based on multi-purpose data sources taking into account some principles (including process data at data source):

- Multi-source statistics
- Multi–purpose data sources
- Pushing computation out
- Use data without sharing
- Layered data organisation
- Modular methodological frameworks

Models of OJA data integration

In order to let alternative data sources gain solid ground in official statistics, [4] propose general framework for integration of these data into the production of official statistics. In the case of using OJA data, three potential scenarios can be foreseen. OJA data need to be pre-processed before they can be linked and integrated to other data sources. Figure 2 is the recommended one, while Figures 1 and 3 are specific cases when the administrative data are impossible to be linked or no survey data are collected.

Scenario 1: Alternative data collection

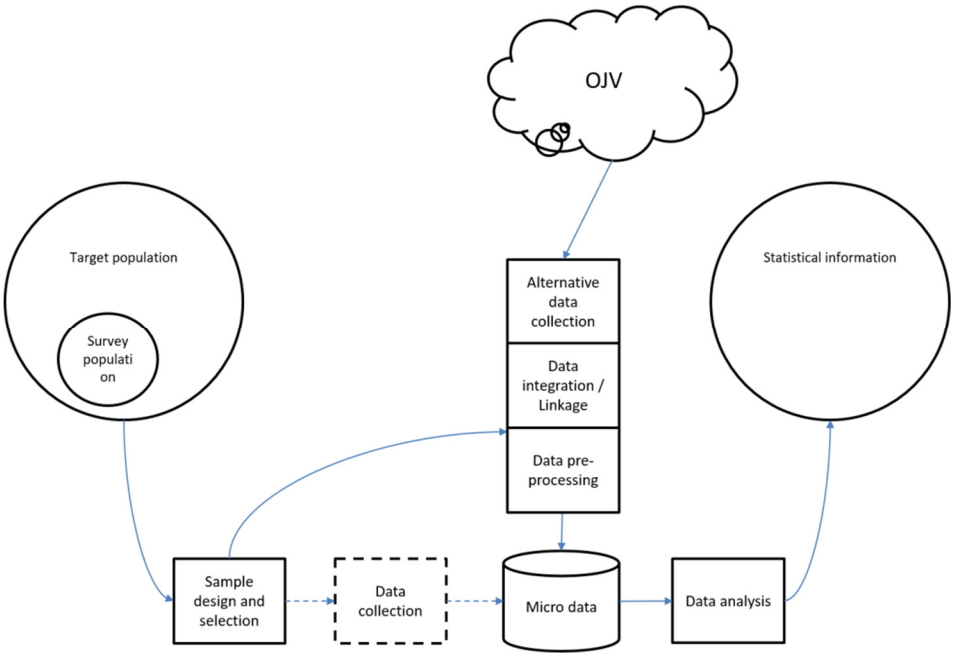


Figure 3: OJA as alternative data collection

Scenario 2: Integrated use

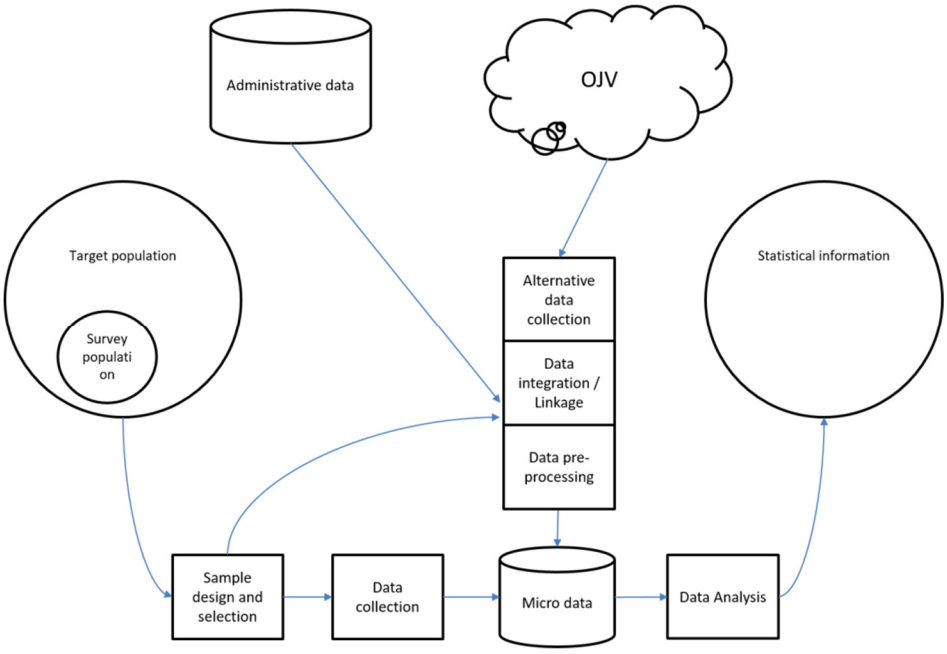


Figure 4: Integrated use of OJA and statistical data

Scenario 3: Substitution of traditional data

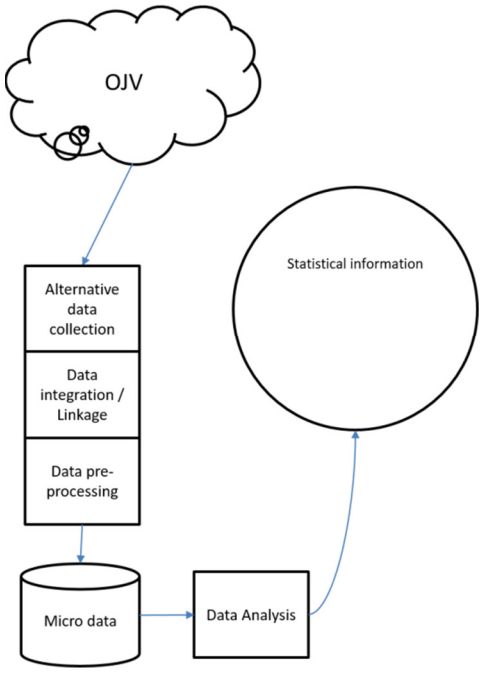


Figure 5: OJA substitute statistical data collection

Common OJA data processing phases

The ESS Statistical Production Reference Architecture (SPRA) [5] uses and introduces a set of key concepts for describing statistical production architecture: Building Blocks, Services, Information Objects, Service Orchestration and Metadata Management. The SPRA model provides a bridge between the Business, Application and Information Architectures. The bridge between both worlds of abstraction is facilitated by the introduction of Services in the sense of a Service-Oriented Architecture (SOA). SOA is an approach applicable to both business and IT. In the business sense, SPRA is about focusing on which services the business requires in order to fulfil its mission, and how the organization of services can be improved.

In the next paragraphs, the main OJA processing phases, building blocks and services considered in the development of an OJA processing system are described.

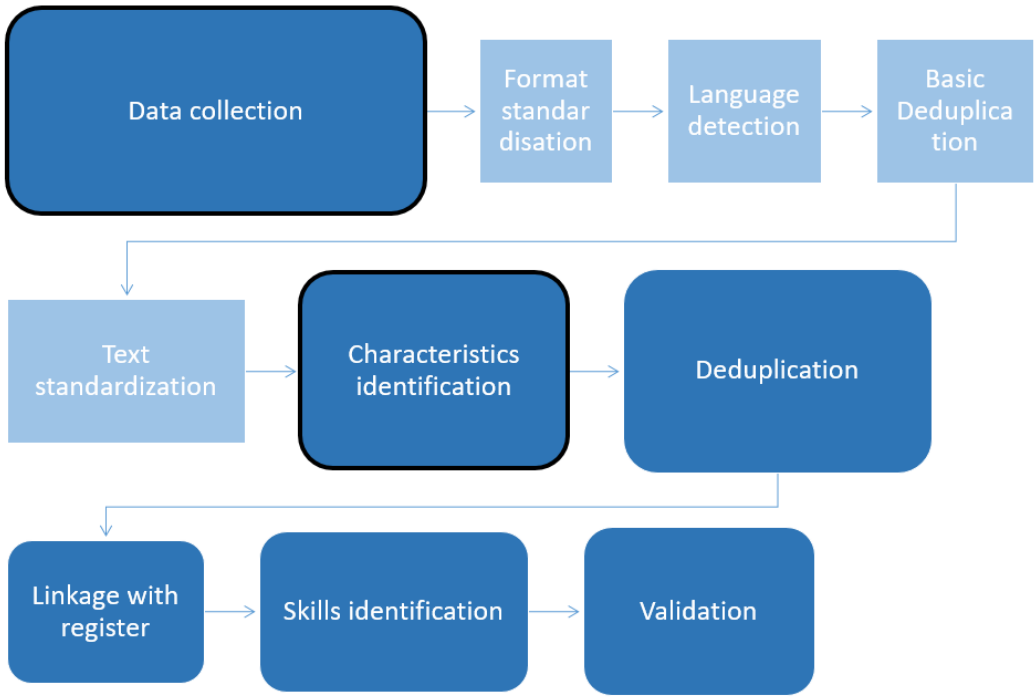


Figure 6: Common ESS OJA data processing phases

OJA Building blocks

The ESS Enterprise Architecture Reference Framework (ESS EARF) [6] defines a set of Building Blocks, which represent (potentially re-usable) components of (IT) capabilities. Conceptually speaking, Building Blocks comprise one or several Business Services. The boundary and specification of a building block should be loosely coupled to its implementation. It should be possible to build blocks in several different ways without affecting the boundary or specification.

In the IT sense, building of blocks is about the integration of various technologies through a common framework, standards and protocols. Building block BB represents potentially re-usable components of IT capability. According to SPRA, Services can be created, either relying on existing IT solutions, or by newly created solutions, or by a mixture of both. The basic Services in turn can be composed to form business architecture.

In this section the main building blocks that implement OJA statistical services of the ESS OJA architecture are described. Some important necessary pre-conditions: where to start, what to consider, phases and main steps are examined.

One approach to describe the management of OJA data is the Knowledge Discovery in Databases (KDD) process for extracting useful knowledge from unstructured data. The KDD process consists of five main steps, as shown by [3] in Figure 5: data source selection, pre-processing, transformation, data mining and machine learning, interpretation/evaluation. To identify related building blocks, this process needs to be adapted to the domain of interest, enhancing one task or step with respect to another.

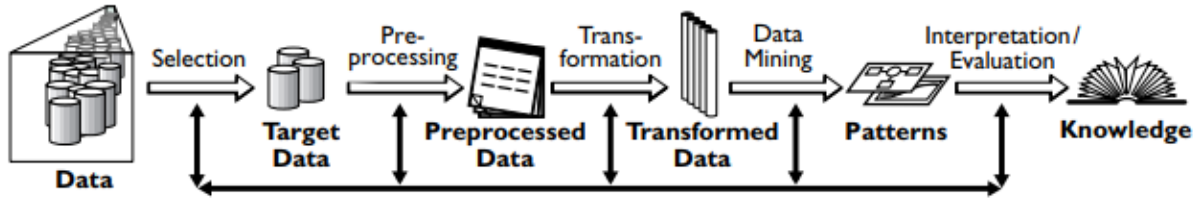


Figure 7: Overview of steps constituting the KDD process

Based on key components of the NIST's general architecture description described in [9] we can define main building blocks of common ESS Reference architecture for processing OJA data.

Table 5: Main OJA building blocks

OJA building block	Short description	Implemented services
Data collection	Data collection refers to the capability of collecting data from various Internet sources in an automated way. It can be run in real time or as a batch process as crawler, scraper or API arranged service.	Crawler Scraper API access Data source selection
Data pre-processing and validation	This building block implements data cleaning methods to remove noise from the data, detect and remove duplicated entries.	Steaming Lemmatisation Stop word removal Text search by string distance comparison Record linkage Deduplication by string comparison Language detection
Data mining and machine learning	This building block implements various data mining and machine learning algorithms to automatically search for patterns of interest depending on the purpose of the analysis: Classification, Language detection, Deduplication, Prediction Regression, Clustering. This building block provides machine learning techniques	Classification Language detection Deduplication by ML Regression

	(ontology based learning, supervised learning and unsupervised learning).	Clustering
Statistical data integration	Record linkage brings together information from two or more records that are believed to belong to the same entity. Record linkage is used to link data from multiple data sources or to find duplicates in a single data source.	SBR linkage Deduplication
Data visualization and analysis	Exploring and visualizing it is an essential step in gaining insight. Common tasks may include visualizing counts and distributions, generating word clouds, and performing distance measures.	
Process orchestrator	Design statistical production processes and workflows in an effective and flexible manner to ensure efficient production as well as easy replication/reusability of the processes. Keep trace of the data processing from beginning to end in the process to ensure the quality valuating.	
Metadata management system	Building block that manages the entire life cycle of reference, structural, and process metadata and logically connects these data. This will form a key building block and prerequisite for standardizing and industrializing statistical production. The building block ensures the availability and integrity of metadata, as well as the suitability of metadata for consistent quality monitoring in the ESS.	
Catalogue for sharing reusable solutions	The primary aim of the catalogue is to support efficient sharing and reuse of services and/or building blocks. It also enables efficient discovery of what is available for reuse. Since many procedures are based on machine learning algorithms, the catalogue provides support for rapid experimentation and iteration and versioning of models and related data sets.	

Primary data storage	Data storage where the primary data from production are stored. The building block supports large amounts of data and the flexible addition of new data sources. It can be split into micro (unit level) data storage and macro (aggregated) data storage as different information (governance) models may apply. The building block includes trusted recovery providing recovery facilities such as restoring from backups in ways that do not compromise security protection.	Semi- and unstructured data storage Structured data storage Recovery service Backup service

Classical NLP (Natural language processing) production lifecycle could be described in four main phases, which continuously evolve due to new data and new features of the data:

- Exploratory data analysis
- Development and testing of ML models and algorithms
- Production
- And continuous improvement

Machine learning is about rapid experimentation and iteration and without keeping track of modelling history; it is not possible to learn much. Effective versioning is needed to keep track of all models developed, as well as their effectiveness, and optimal combination of input and output parameters, algorithm choice and applied architecture. Effective model versioning supports:

- Finding the best model
- Failure tolerance
- Increased complexity and file dependencies
- Gradual, staged deployment

According to the identified building blocks in Table 2, Figure 8 proposes ESS Reference OJA architecture, following also some principles of future ESS Web intelligence hub as defined in [7]:

- Serving national and European needs
- Modular structure
- Defined processes and products to be guaranteed
- Priority to working together, possibility to act individually
- Programs should be open source
- Transparency as much as possible
- Common used processes should be certified and audible
- Lineage of data and processes
- Intermediate products usable by all partners

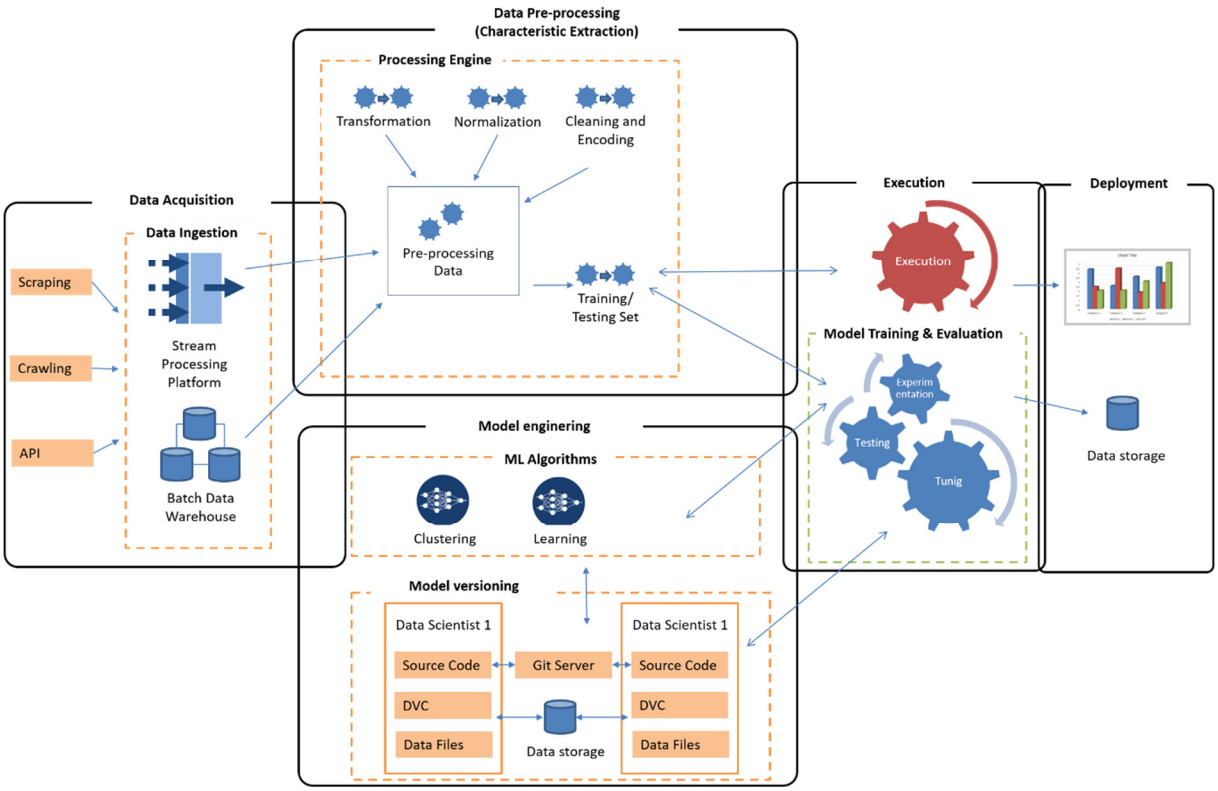


Figure 8: ESS Reference architecture for processing OJA data

OJA Services

Table 6: Main OJA Services

OJA service	Short description
Data source selection	As the world of OJA is highly dynamic, these processes should be repeated and continuously improved on a regular basis. This service identifies and selects data sources according to methodology for data source classification and ranking in order to secure uninterrupted data access.
Scraping	Scraping is a data collection activity that extracts meaningful data from web pages. Only predefined information is collected from the source by using software (scraper) developed according to the data sources' (webpage) known data structure.
Crawling	Crawling is a data collection activity that automatically searches and indexes complete data sources' content. It

	iteratively follows the structure of the data source.
API access	API access is a data collection activity that utilizes a software component exposed by the source owner.
Text data pre-processing & feature extraction	This service turns the raw text extracted from data sources into structured data using text mining and NLP tools. Text pre-processing consists of a number of steps, any number of which may or not apply to a given task, but generally fall under the broad categories of tokenization, normalization, and substitution. Every structured data item extracted from the data source is mapped onto reference taxonomy to achieve a common data representation.
Language detection	Before word and sentence segmentation, character normalization or lemmatization can occur, the language of the source document must be determined. Language detection service computationally determines the language of a given piece of text data.
Ads deduplication	The same employer can publish the same job vacancy to different websites at different times. Deduplication is an activity for finding records in datasets belonging to the same entity. In OJA case, we try to identify and de-duplicate the same OJA adverts posted on different web portals.
Identify characteristics	Characteristic identification (text classification) assigning tags or categories to text according to its content.
Generate/maintain ontologies	Ontology is a formal description of knowledge as a set of concepts within a domain and the relationships that hold between them. To enable such a description, we need to formally specify components such as individuals (instances of objects), classes, attributes and relations as well as restrictions, rules and axioms. As a result, ontologies service can also add new knowledge about the domain.
Data validation and quality	The objective of the data validation is to identify errors and issues in data in order to fix them and give quality assessment. It involves data management, analysis, quality control and assurance. Data quality assessment includes modelling of data completeness, accuracy, timelines, consistency, and deduplication.
Data Exploration & Visualization	

Record matching	Link the OJA to the Business Register for deducing variables such as work unit.
Data standardisation (see figure 9)	

The ESS Enterprise Architecture Reference Framework (ESS EARF) [6] defines a business capability as ability that organisation, person or system possesses. Business capabilities serves as a basis for developing business architectures. They represent what business does and are similar to business functions as defined as Common statistical production architecture standard (CSPA) which defines process/activity oriented operational context. Figure 7 represents OJA business functions and processes.

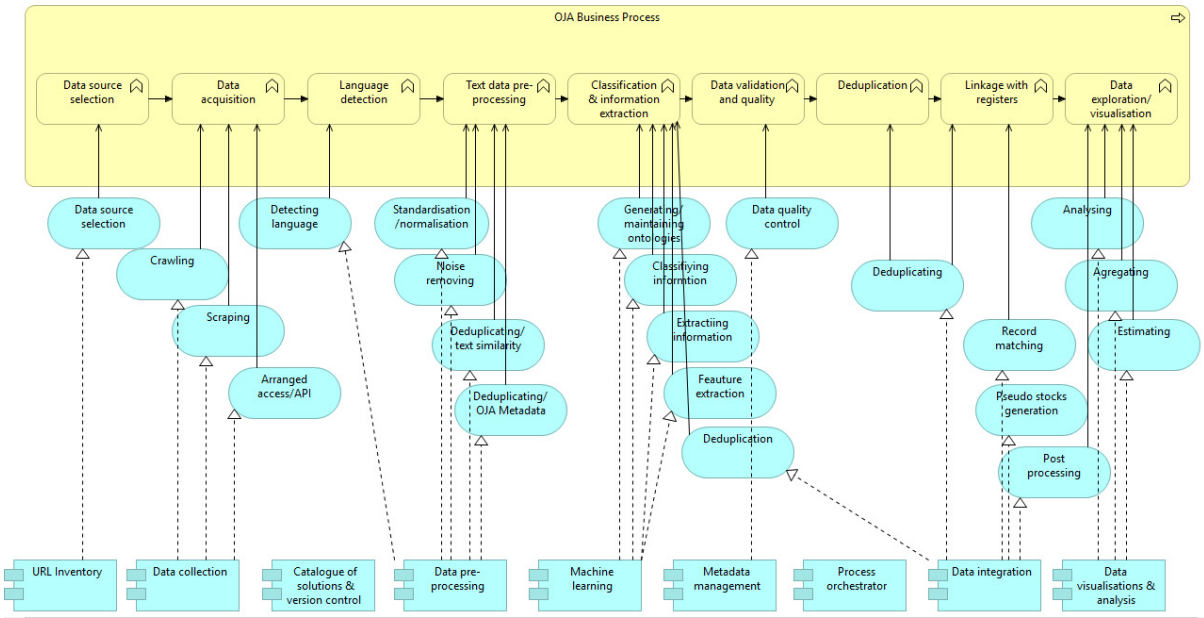


Figure 9: OJA application architecture

ESS OJA implementation scenarios

From a conceptual point of view, the architecture to process OJA data should appear as in Figure 6. The goal of such architecture is to collect OJA data from Internet sources, extract useful information from them, decouple the specificity and complexity of OJA data sources from the statistical analysis and promote collaborated development and sharing of solutions within the ESS. From this perspective, the proposed conceptual architecture consists of (at least) three main building blocks: data collection, pre-processing and characteristics extraction, and statistical processing and analysis.

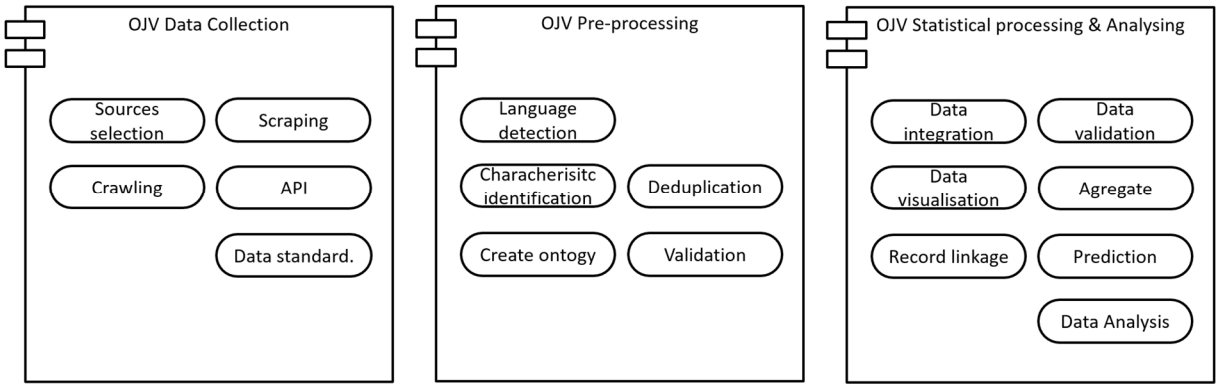


Figure 10: High-level building blocks that implement main OJA processing services

The SPRA acts like a standard and provides guidance for developing Information System Architectures affecting the whole of the statistical production chain.

The ESS EARF also defines possible consolidation approaches for the Building Blocks, Services and Information Objects in the SPRA:

- Autonomous
- Interoperable
- Replicated
- Shared

There are two potential scenarios for statistical institutes to implement the OJA production architecture. These scenarios can be supported simultaneously with the architecture and are thus not mutually exclusive.

Scenario 1 – ESS OJA implementation scenario: Replicated

In this scenario, the ESS member locally implements autonomous OJA data production system that is interoperable with the ESS OJA Production System enabling it to utilize a selection of the shared statistical services.

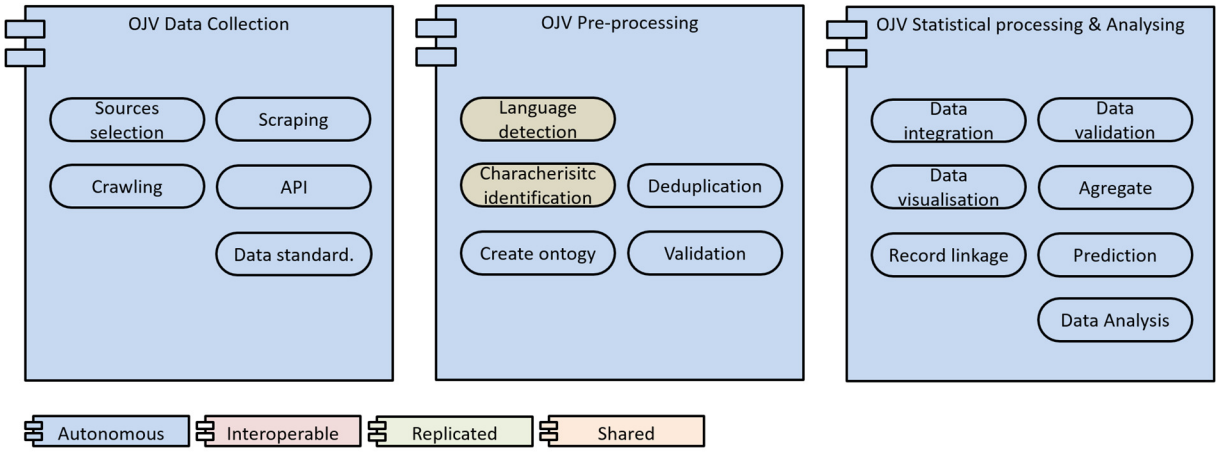


Figure 11: ESS OJA implementation scenario: Autonomous

Common ESS Services can be replicated locally and integrated with locally developed Services. The statistical institute implements a particular replicated building block as “Black box” and provides country-specific input data and metadata.

Scenario 2 – ESS OJA implementation scenario: Shared

Starting from the interests and building strengths of all parties (Member States and Eurostat), there is a ground for collaborative development of a shared system for producing data from OJA. Such a system should cover the need of all Member States and all official languages gather OJA data from wide range of preselected websites and portals, cover all stages of data processing: data collection, data pre-processing, characteristics identification, information extraction and analysis.

In this scenario, the ESS member proactively participates in the development processes as well as provides country-specific input data, metadata and algorithms in order to secure sustainable technical solution for the production of high quality, complete and comparable OJA data. An important aspect of this collaboration is also organisation of work, including possible roles and responsibilities of each party as well as governance structure <to be described>.

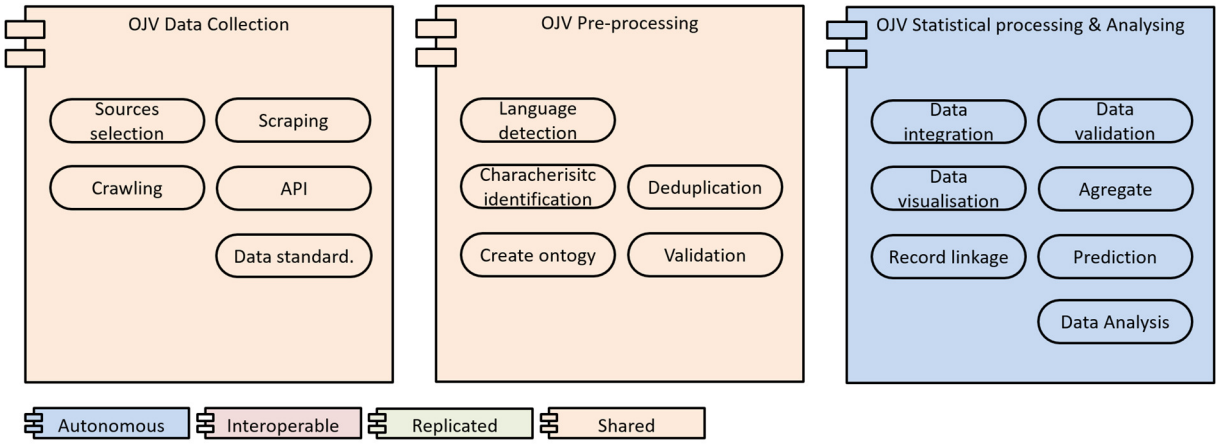


Figure 12: ESS OJA implementation scenario: Shared

According to SPRA, no matter whether a statistical institute chooses to implement one or the other scenario, a gradual implementation remains possible. The effort of making the statistical services available by wrapping existing applications or building new ones is significant. In addition, the design and implementation of the actual statistical processes is a significant change effort. For these reasons, the suggested architecture also takes into account the NSI’s specific production systems of those statistical institutes that cannot find positive business cases for moving the production of specific stable statistics to the common production system.

Occupation classification

There are several different classifications that can be extracted from the job vacancies. The most common classification for occupations is ISCO (International Standard Classification of Occupations) or ESCO (European Standard Classification of Occupations)[6].

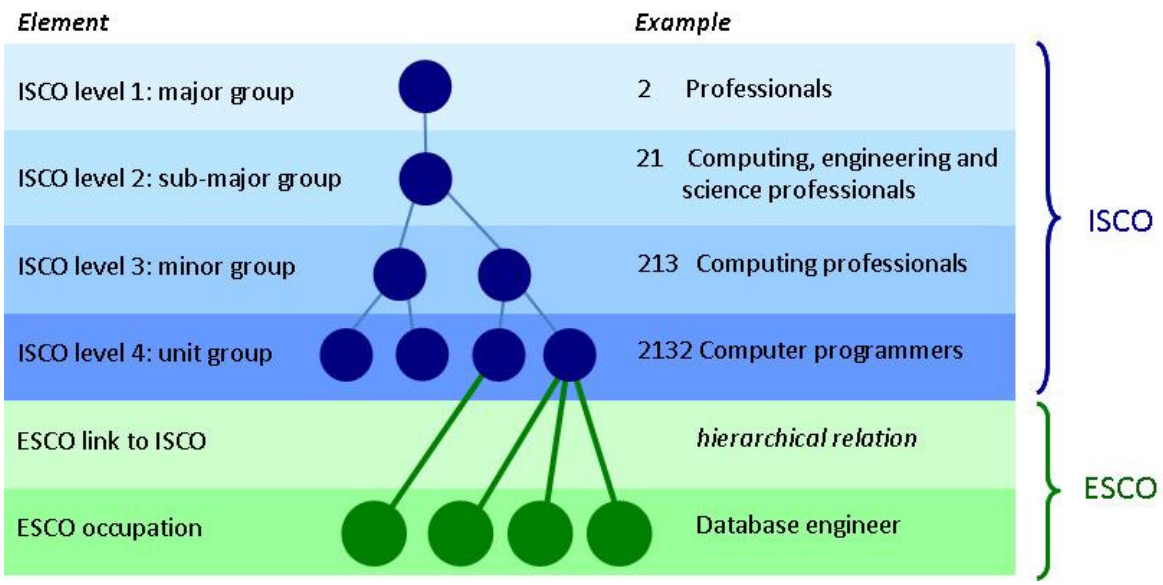


Figure 13: ESCO vs. ISCO - mapping the occupation level

ESCO classification includes 2,942 occupations (as of 5 September 2019) divided into 10 groups[7]:

- 0 Armed forces occupations
- 1 Managers
- 2 Professionals
- 3 Technicians and associate professionals
- 4 Clerical support workers
- 5 Service and sales workers
- 6 Skilled agricultural, forestry and fishery workers
- 7 Craft and related trades workers
- 8 Plant and machine operators and assemblers
- 9 Elementary occupations

Each occupation contains one preferred term and any number of non-preferred terms and hidden terms in each of the ESCO languages. Each occupation has its own profile, which contains an explanation of the occupation in the form of description, scope note and definition. They also list

[6] https://ec.europa.eu/esco/portal/escopedia/ESCO_in_statistics, as of 5 September 2019
[7] https://ec.europa.eu/esco/portal/occupation, as of 5 September 2019

knowledge, skills and competences needed for each occupation. In full view of the ESCO classification it includes:[8]

- Description
- Alternative label
- Regulatory aspect
- ISCO-08 code
- Definition
- Scope notes
- Hierarchy
- Narrower occupations
- Essential skills and competences
- Essential knowledge
- Optional skills and competences
- Optional knowledge
- Status
- Replaces
- Replaced by
- Concept URI

This wide scope of the definition of each occupation does allow for a more refined narrower classification.A snapshot of the 4-level hierarchy according to the CEDEFOP dataset is presented in Figure 13.

	idesco_level_4	esco_level_4	idesco_level_3	esco_level_3	idesco_level_2	esco_level_2	idesco_level_1	esco_level_1
0	2634	Psychologists	263	Social and religious professionals	26	Legal, social and cultural professionals	2	Professionals
1	2352	Special needs teachers	235	Other teaching professionals	23	Teaching professionals	2	Professionals
2	4321	Stock clerks	432	Material-recording and transport clerks	43	Numerical and material recording clerks	4	Clerical support workers
3	7131	Painters and related workers	713	Painters, building structure cleaners and rela...	71	Building and related trades workers, excluding...	7	Craft and related trades workers
4	2512	Software developers	251	Software and applications developers and analysts	25	Information and communications technology prof...	2	Professionals

Figure 14: ESCO different levels representation in CEDEFOP dataset

There are two common methods to map the occupation with the classification code:

1) Textmining

[8] https://ec.europa.eu/esco/portal/occupation, as of 5 September 2019

2) Machine learning

Both methods have some common tasks to do before the matching algorithm is used. It was presented in Figure 3.

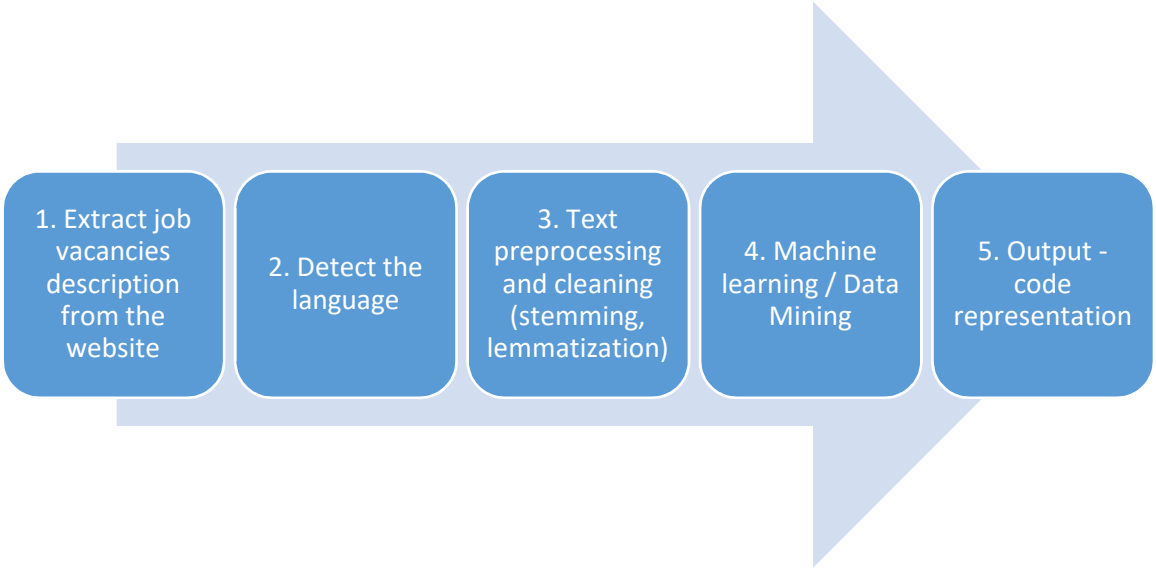


Figure 15: General steps in the process of ESCO code detection

To start the process of classifying an occupation, it is necessary to extract relevant data from the website. Usually it is a short job description. The second step is to detect the language. It is important to match the language with the job offer. Otherwise, the occupation code will not match correctly. The third step is to clean the data – it includes stop words removal, stemming and lemmatization, briefly described in the previous section of this methodological framework.

The fourth step is to use the machine learning or text mining algorithms to match the description of the job offer to the proper ESCO code. In the first case, a training dataset has to be created and defined for each language separately. The last step is to provide the code representation of the occupation for the detected job vacancy, in hierarchical form, as presented in Figure 13.

According to the architectural building blocks used in this methodological framework, the following building blocks are involved in the occupation classification process:

- Data collection
- Data pre-processing and quality
- Data mining and machine learning

References

[1] Web scraping / Job vacancies Deliverable 1.1, Strategy for ongoing engagement (SGA-2), Last accessed August 2019 at:
https://webgate.ec.europa.eu/fpfis/mwikis/essnetbigdata/images/e/e0/WP1_SGA2_Deliverable_1_1_1.0docx.pdf

[2]Web scraping / Job vacancies Deliverable 2.2 Final Technical Report (SGA-2), Last accessed August 2019 at:
https://webgate.ec.europa.eu/fpfis/mwikis/essnetbigdata/images/5/5e/SGA2_WP1_Deliverable_2_2_main_report_with_annexes_final.pdf

[3] Fayyad, U., Piatetsky-Shapiro, G. and Smyth, P., 'The KDD process for extracting useful knowledge from volumes of data', Commun. ACM, Vol. 39, No 11, 1996, pp. 27–34., Last accessed March 2020 at: https://sceweb.uhcl.edu/boetticher/ML_DataMining/p27-fayyad.pdf

[4] Diego Zardetto, 'The implication of Big Data for Official Statistics'. Last accessed August 2019 at:
https://circabc.europa.eu/sd/a/c35f00b3-3103-4873-a5a8-68386822af03/DAY%201_ITEM%202_The%20implication%20of%20big%20data.pdf

[5] ESS Statistical Production Reference Architecture, Last accessed August 2019 at:
https://ec.europa.eu/eurostat/cros/system/files/Statistical%20production%20reference%20architecture%20v04.pdf

[6] ESS Enterprise Architecture Reference Framework, Last accessed August 2019 at:
https://ec.europa.eu/eurostat/cros/content/ess-enterprise-architecture-reference-framework_en

[7] Trusted Smart Statistics Strategy, Last accessed Marc 2020 at:
https://webgate.ec.europa.eu/fpfis/mwikis/essnetbigdata/images/1/1f/2019_06_03_BD_ESSnet_WP_B_TSS.pdf

[8] Fabio Ricciato, Towards a Reference Methodological Framework for processing MNO data for Official Statistics, Last accessed August 2019 at: http://www.15th-tourism-stats-forum.com/pdf/Papers/S3/3_1_A_Reference_Methodological_Framework_for_processing_mobile_network_operatordata_for_official_statistics.pdf

[9] NIST Big Data Interoperability Framework: Volume 5, Architectures White Paper Survey, Last accessed August 2019 at: https://bigdatawg.nist.gov/_uploadfiles/NIST.SP.1500-5.pdf

[10] Proxy in statistics:
Upton, G., Cook,I. (2002) *Oxford Dictionary of Statistics*. OUP ISBN 978-0-19-954145-4

[11] Regulation (EC) No 453/2008 , Last accessed Marc 2020 at: https://eur-lex.europa.eu/LexUriServ/LexUriServ.do?uri=OJ:L:2008:145:0234:0237:EN:PDF

[12] Proceeding of the 1st and 2nd International Workshops on Methodologies for Job Vacancy Statistics, Eurostat Methodologies and Working papers, 162 pp ISSN 1977-0375, Last accessed Marc 2020 at:
https://ec.europa.eu/eurostat/web/products-statistical-working-papers/-/KS-RA-10-027

[13] Saucy, F. Analysis of X28 AG test files, internal report for WPB OJA presented at the ESSnet BigData NDL meeting at Thessaloniki, March 2019 (available on request).

[14] Martina Rengers, Jakob de Lazzer, CEDEFOP online job vacancies file for Germany (DE), WPB OJA, sept. 2019, 7 pp., https://webgate.ec.europa.eu/fpfis/mwikis/essnetbigdata/images/1/19/WPB_Deliverable_B1_Interim_technical_report_Annex_DE_CEDEFOP_exploratory_analysis.pdf

[15] Saucy, F., Schmassmann, S. & Columbano, A. Analyses of 1st CEDEFOP DATASET (June 2019), Aug. 2019, 41 pp., https://webgate.ec.europa.eu/fpfis/mwikis/essnetbigdata/images/c/cc/WPB_Deliverable_B1_Interim_technical_report_Annex_CH_Analyses_of_1st_CEDEFOP_DATA_%28July-Oct._2019%29.pdf